

The Many and The One: What Composite Traits Are and Why They Matter

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Abstract

The trait is the fundamental unit of biology. To study evolution or make comparisons across taxa, researchers must identify the same trait in different individuals. Yet because organisms are integrated wholes, dividing them into constituent traits is no trivial task. In this article, we discuss a class of traits that pose unique challenges: composite traits. A composite trait is a trait built out of constituent traits, where the composite carries biological significance beyond that of its individual constituents. We show that composite traits are valuable to the biological sciences but pose deep challenges, especially to comparative biology.

1. Introduction

In 2022, astronomers reported the discovery of the most distant star ever observed. Initially detected through a distortion in the path of light (gravitational lensing), the star was later captured in images by the James Webb Space Telescope. Named Earendel—“Morning Star” in Old English—it shines brilliantly against the red haze of its host galaxy, the Sunrise Arc, offering a glimpse into the universe’s first billion years. However, what was once thought to be a single star may, in fact, be a composite—that is, a cluster of stars. Unlike constellations, which are patterns we project onto the night sky and are often composed of isolated heterogeneous celestial light sources, star clusters are proximate and gravitationally bound.

Given the extreme distances involved, distinguishing between a single star and a composite is far from straightforward. But sometimes, even for objects right in front of us, it is difficult to individuate them as singular items or as composites. This is especially a problem for biology. Here, the basic biological unit is the *trait*¹ and assumptions that part of a phenotype of interest is a singular trait rather than a *composite* can have important implications for how we understand the development, ecology, and evolution of organisms. For example, failing to recognize a composite can mislead us into searching for a singular adaptive explanation in the absence of a unitary selective history.

¹ We are using ‘trait’ and ‘character’ interchangeably, although we acknowledge that these are sometimes distinguished in some subdisciplines.

Biologists have explicitly recognized the importance of composite traits across a variety of fields (e.g., Riska 1989; Badyaev et al. 2001; Opedal et al. 2017; Chomicki et al. 2024). One example is herkogamy—the spatial separation between the anthers and the stigmas within flowers. Herkogamy is considered a composite functional trait in plant reproductive biology, promoting outcrossing and avoiding interference between male and female functions (Opedal et al. 2017). However, not all composite traits need to involve components that are in proximity: the trait of bipedalism, for example, is arguably a composite of widespread skeletal features, including those in the pelvis, spine, foramen magnum, feet, and femur (Harcourt-Smith 2007). Other candidate composites may consist of significantly heterogeneous elements, such as language, which involves a cluster of anatomical, sensorimotor, cognitive, and sociocultural traits (Hauser, Chomsky and Fitch 2002; Arnon et al. 2025).

While composite traits have been recognized and discussed in biological literature, a philosophical articulation of what composite traits are, how they are formed, and the challenges they pose for evolutionary and comparative investigations remains largely neglected. Meneganzin and Currie (2025) drew philosophical attention to this issue in the context of anthropological studies, focusing on how the neglect of trait complexes² impacts

² In this paper we use ‘composite trait’ interchangeably with ‘trait complex’, with the key difference that the former more explicitly foregrounds questions of individuation concerning “the one” in light of “the many” (the title of this paper). However, we acknowledge that

the evolutionary significance of common empirical practices, especially for complex life-history stages such as childhood. This paper aims to foreground the broader problem of composite trait individuation by analyzing the nature and role of composite traits across different areas of evolutionary biology research, showcasing their importance and the challenges they pose.

With Earendel, a critical question is whether the light emanates from a single star or from multiple objects. In biology, we need to ask how single and composite traits are distinguished and how the composite itself is characterized. Since biological characters span a range of categories, we will consider three examples of candidate composite traits currently discussed in the empirical literature. These examples—plumage coloration in birds (a property of a morphological structure), the trapping mechanism in pitcher plants (an activity), and hominin childhood (a life history stage)—highlight the multiple ways in which composites can be realized.

The problem of composite trait individuation comes to the fore when we ask comparative questions across taxa. Do meerkats have childhoods, for instance? Childhood is a human-based composite trait (Meneganzin and Currie 2025), but what are the traits that

stricter non-overlapping usages are also possible, such as complexes and composites standing on a spectrum of progressive integration. For reasons of space, we do not delve into this finer-grained articulation here.

compose it? To test whether meerkats (or other species) have childhoods—allowing for the possibility of reevaluating the unique status of a character and making similar comparative questions beyond humans tractable—we need to know (1) which traits are essential to the composite (as opposed to being merely correlated with it) and (2) how to identify instances of the “same” composite across taxa. Thus, for it to be possible for a composite trait (like childhood) to occur in other taxa, we need to have a way of generalizing the minimal requirements for realizing the composite.

In this paper, we will consider the nature of composite traits, the investigative strategies they support, and what challenges they carry for comparative biology.

2. Composite Traits

One of the foundational issues in biology is the principled individuation of its basic unit of analysis: the trait. Traits need to be singled out and tracked—in diverse organisms through developmental and evolutionary time, across taxa, and in distinct environmental settings—for empirical questions to gain traction. Evolution, for instance, is about trait changes over generations. Any assertion that evolution occurs requires identifying the *same* trait in different organisms. The theoretical challenges presupposed by such acts of delimitation are collectively referred to in philosophy as the thorny problem of *trait individuation* (DiFrisco and Ramsey 2023; Figdor 2024; Meneganzin et al. 2024; Cortés-García et al. 2024; Fábregas-Tejeda 2025; DiFrisco 2025).

Trait individuation becomes a salient issue insofar as it raises the possibility that traits can, among other things, be ill-delimited or misidentified, thereby undermining their potential to fulfill investigative goals. There is a diversity of approaches to trait individuation and different answers to the question of what constitutes a legitimate trait—what has been dubbed the “justification” step of the trait individuation problem, alongside issues of how traits are described and measured (Meneganzin et al. 2024). It is within the justificatory arena that we can assess the promises and limits of functionalist and structuralist approaches to trait individuation (DiFrisco 2025; DiFrisco and Ramsey 2023; Fábregas-Tejeda 2025). Roughly, the functionalist approach individuates traits based on the functions they carry, where these functions are endowments of an evolutionary response to natural selection (e.g., Arnold 1983; Brandon 1999; Orr 2000). Structuralists, by contrast, take it as a requirement to focus on how traits develop. They individuate them based on the causal mechanisms that build traits during developmental processes (e.g., Wagner 2007; Wagner 2014; DiFrisco et al. 2020, 2022).

Whether the functional or the structuralist route is adopted—or a hybrid account is devised—it seems that a minimal commitment to variational (quasi-) independence, broadly construed, is hard to escape: the ability of a candidate trait to change relatively independently from others, either through development or by being singled out as a specific target of a selective process, is an important part of the justification for identifying something as a trait (Meneganzin et al. 2024). For the purposes of this paper, we will assume this minimal

commitment while backgrounding the richer debate on trait individuation to focus instead on composite traits: what they are and what role they play in the evolutionary sciences.

When astronomers proposed that Earendel is a star composite, this undermined its status as a star. Earendel cannot be both a star and a star composite. With composite traits, by contrast, we argue that something can be a *bona fide* (composite) trait and also be composed of traits³. This can occur in different ways. One obvious way is to recognize that composite traits and their component traits occur at different levels of biological organization.

To see how new traits emerge at higher levels of organization, take the example of sex ratio. Sex ratio is the relative number of females to males in a population of sexual organisms. It can be understood as a population-level composite trait⁴ made up of organismal (thus, lower-level) individual traits—their sex, in this case—and their relative proportions in the population. No conflict arises in calling both the populational sex ratio and the organismal sex

³ As such, it would be a mistake to read this claim as undermining the project of trait individuation rather than expanding it. Because the “traithood” of individuated entities can be multiply realized across levels of organization and categories of characters, there is no contradiction in attributing distinctive traithood to both individual traits and composites. We are grateful to Alejandro Fábregas-Tejeda for drawing attention to this point.

⁴ Research applying optimization models to sex ratio and investigating its evolution treats sex ratio as a trait subject to natural selection (e.g., Charnov 1982).

‘traits’, thereby opening the possibility for nested traits across levels. Furthermore, the higher-level trait—sex ratio—carries biological significance that goes beyond the significance of the component traits: sex ratio is linked to an array of ecological and evolutionary factors, such as the population’s growth rate.

Just as sex ratio is not a mere insignificant collection of traits, but is a population-level composite that carries biological consequences, so organism-level composite traits are collections of (one or more) traits that have biological significance transcending the significance of their component traits. Crucially, this biological significance is realized by means of patterned relationships among the traits, such as the traits forming a functional whole or being integrated in nonrandom ways. In the case of sex ratio, for instance, it has been shown why a 1:1 ratio between the sexes in a population constitutes an evolutionary stable strategy for most species (Fisher 1930).

To see the difference between true composites and mere collections of traits, consider another trait of organisms: their mass. While sex ratio is a biologically significant composite, population mass is not a composite in our sense. While the masses of individuals are important, population mass, except in some unusual cases, will not carry additional biological significance. It doesn’t form a new trait with unique roles to play. Our contention is that some conjunctions of traits are like mass: they don’t form significant traits of their own, whereas others are like sex ratio, in which the composite is itself an important trait. We will thus

restrict ‘composite’ to trait collections that stand in patterned relations to one another and, as such, have biological significance transcending the significance of their components.

3. Three examples of composite traits

Here, we consider three examples of proposed composite traits drawn from biology and anthropology. The diversity of the examples highlights the disciplinary reach of the problem and the multiple ways in which composite traits can be realized. We show why they are true composites and not mere collections of traits, before illustrating the theoretical and investigative approaches they enable and the challenges they entail.

3.1 Plumage coloration

Among studies of sexual ornaments, plumage coloration is one of the most extensively investigated phenotypes. Carotenoid-based plumage color is recognized as a distinct type of ornament (alongside melanin- and structure-based ornaments). Crucially, it has been suggested that research on condition-dependent sexual ornaments has been muddled by the lack of recognition that the target trait—a *property* of a morphological structure (the feathers)—is a composite trait. At least four components of this composite have been identified: pigment elaboration, patch area, pigment symmetry, and patch-area symmetry (Badyaev and Hill 1999; Badyaev et al. 2001).

What this entails is that “plumage coloration” is not a simple trait but is instead a composite carrying significance beyond the significance of its components. The components, in turn, are individuated by means of developmental independence: pigment elaboration, for instance, depends on the type and quantity of pigments available to receptive follicles during feather growth, while patch area varies according to the number of follicle cells that are receptive to carotenoid pigments during feather growth (Badyaev et al. 2001).

The independence of the traits forming the composite is supported by empirical studies. For instance, in male house finches, not only do the component traits show developmental independence, but they also partake in selective processes in different ways (pairing, viability, and fecundity), with their variation bearing different functional and fitness consequences (Badyaev et al. 2001). As Badyaev and colleagues note, the components flexibly combine, “ultimately producing a composite and dynamic secondary sexual trait” (Badyaev 2001, p. 229). Plumage coloration thereby takes on distinct biological significance across comparative contexts, conveying different information about the physiological and behavioral condition of its bearers.

3.2 Tropical pitcher plants

Tropical pitcher plants (genus *Nepenthes*) capture insects using pitfall traps—modified leaves with slippery inner surfaces and a lid-like operculum. In most species, the lid contributes little to prey capture. In *Nepenthes gracilis*, however, it performs a distinctive role: raindrops

striking the lid cause it to snap downward like a springboard, catapulting insects into the digestive fluid. A recent study reported that *Nepenthes pervillei*, a species endemic to the Seychelles, exhibits a comparable springboard mechanism (Chomicki et al. 2024). These two species are separated by several thousand kilometers on opposite sides of the Indian Ocean, prompting the question of what underlies the evolution of their similarly adaptive trapping mechanism.

Here, the focal unit of analysis (springboard trapping) is an *activity* generated by a composite: (i) a horizontally oriented lid, (ii) a slippery underside, and (iii) a stiff, spring-responsive lid (Chomicki et al. 2024). These component traits show considerable variation across *Nepenthes* species. Their ancestral state and phylogenetic analyses speak in favor of their independent evolution, thus making a strong case for their variational independence. The trapping function is a novel, synergistic function bearing independent biological significance: no single component trait suffices on its own. The emergent function is realized thanks to a patterned relation among the components, justifying the treatment of the springboard mechanism as a composite trait.

3.3 Human childhood

Childhood is often described as one of the distinctive, derived traits of the genus *Homo* (Bogin 1997; Bogin 2020), with the particularly extended childhood of *Homo sapiens* considered unique to our species (Leigh and Park 1998). But what is childhood? Because multiple traits

contribute to defining it—including the growth rates at which different body parts develop, feeding behavior, and the development of cognitive and social skills—it is best understood as a composite rather than a single trait (Meneganzin and Currie 2025). The composite in this case constitutes a *life-history stage*.

What makes the case of childhood particularly interesting—and challenging to define—is that, although several diverse traits are related to childhood, there is no consensus on which of them is indispensable to childhood. That is, which traits are part of the childhood composite, and which are merely correlated or associated with childhood? Indeed, there is variation in how childhood is characterized across cultures and anthropological subdisciplines. Some view childhood as a “social artifact,” while others ground it in biology (Meehan and Crittenden, 2016). Within the biological characterizations, there is debate about the relative importance of different markers, including postweaning dependence (Bogin 1997; Gurven and Walker 2006), a decelerated rate of skeletal growth (Bogin 1997; Bogin and Varea 2020), hormonal regulatory mechanisms (Bernstein 2016), behavioral and cognitive traits, and emotional aptitudes (Hrdy and Burkhardt 2020).

Even when considering postweaning dependence alone, one must invoke subcomponent traits as diverse as physiological, nutritional, and social-behavioral features. While these debates and complexities make it challenging to pinpoint the components defining childhood, it is clear that childhood is a genuine composite trait. Crucially, several of

the presumed component traits, such as the timing of weaning and specific patterns of growth, show clear variational independence across primate lineages.

Meneganzin and Currie (2025), considering the broad phylogenetic scope of *Homo*, referred to childhood as a “trait complex,” suggesting that this complex converged at some point in the human evolutionary lineage—thus realizing a true composite trait. They suggested that the broader tapestry of integration and independence among component traits has itself evolved, supporting adaptability under different social and environmental conditions, such that whenever childhood component traits were recognizable in ancestral taxa, childhood as a composite trait likely carried distinct biological significance.

As we will see, composites like childhood are important for understanding the evolution of *Homo*, but they pose challenges when we make comparisons across taxa. Identifying composites in other species requires making commitments about which constituent traits are essential to individuate the composite and which are accidental. We will first consider the benefits of composites before turning to the challenges they pose.

4. The Complexities and Benefits of Composite Traits for Evolutionary Sciences

Recognizing composites as such, rather than as single traits or as arbitrary collections of traits, matters. The recognition of composite traits brings into view specific evolutionary questions and investigative strategies and supports distinctive forms of comparative analysis among

different bearers of the composite. It can be argued that composite and single traits coordinate non-overlapping problem agendas (*sensu* Love 2008)—that is, a set of interrelated questions and associated criteria for how to best address them. Asking what makes a composite unique to a phylogenetic group, for instance, requires articulating what it means to trace the unique state of a unit that, albeit showing a degree of cohesiveness, is made of other subunits. Consequently, what counts as an adequate answer differs for single and composite traits. Composites raise genuinely novel questions. For example, which processes and mechanisms ground the patterned relationship among component traits? How do the components come together to form a whole? Crucially, these questions—and the standards for answering them—only become visible once the composite is properly recognized and characterized.

Consider plumage coloration again. As we have seen, the composite functions as a sexual ornament defined by (at least) four components. Because condition-dependent sexual traits are often subject to strong directional selection (Price et al. 1993), understanding evolutionary responses in a single trait differs importantly from understanding them in a composite. In some cases, a single trait might be characterized using a single dimension. Composites, by contrast, are necessarily multidimensional. Recognizing a composite as such, therefore, introduces several layers of complexity (cf. Ramsey and Meneganzin 2025).

First, there is *variational complexity*: when more than one component is involved, the composite trait can vary in multiple diverse ways, reflecting the contribution of each component. This variational complexity demands careful and adequate characterization.

Second, it entails *metrological complexity*, requiring more targeted and component-sensitive measurement strategies. As Badyaev and colleagues have argued, a generalized measure of overall condition, without any further specification, can obscure the heterogeneous dynamics through which the ornament is realized (Badyaev et al. 2001). Third, there is *explanatory complexity*: how can the observed pattern of variation be explained in terms of the underlying properties, processes, and mechanisms?

If the evolution of sexual ornaments is analyzed in terms of the relative importance of “internal” and “external” processes, such as the developmental predispositions that may be required for sexual selection to favor elaboration (Price et al. 1993), then a single tradeoff is unlikely to provide a satisfactory explanation of a composite trait. Multiple selective pressures shape trait elaboration, with both production costs and their toleration varying across environments. Consequently, the biological significance of plumage coloration—and the information it conveys about its carriers—differs markedly when it is analyzed as a single trait rather than as a composite.

When the composite is a function—such as the trapping mechanism of pitcher plants—explanatory complexity highlights different evolutionary scenarios and ways to discriminate among them. Because the higher-level function depends on the coordinated interaction of three components, treating the trapping mechanism as a composite enabled Chomicki and colleagues (2024) to formulate alternative scenarios: directional selection acting independently on each component, “spontaneous coincidence” (p. 1), or a mixed pathway

combining both processes. A combination of phylogenetic analyses, field experiments, microscopy, and chemical analyses was used to conclude that in two *Nepenthes* species, the trapping mechanism evolved independently through a spontaneous coincidence, drawing together components of the composite that converged through stochastic phenotypic plasticity. Examples like this highlight the insight that composite trait identification can have for understanding the emergence of evolutionary innovations.

5. The Problem of Composite Traits for Comparative Biology

Composite traits inspire coordinated research agendas but also present considerable challenges. One such challenge is distinguishing *core* components of a composite from merely correlated ones—that is, separating the components that *define* the trait from those that may be carried along for various evolutionary reasons.

In the context of the pitcher plant, delimiting the core of the composite is relatively straightforward: if no subset of the components is sufficient to realize the function, all three traits are considered part of the core. But consider a life-history stage like childhood. How would we determine which component traits fundamentally individuate childhood? This question is particularly pressing given the strong claims about the derived character of childhood in the genus *Homo*. Childhood is argued both to be a novel life-history stage in *Homo* (Bogin 1997; Bogin 2020) and to be disproportionately prolonged in the *Homo sapiens*

lineage—thus carrying an additional degree of uniqueness within a narrower phylogenetic scope (Leigh and Park 1998).

The first claim requires establishing the novelty of the composite in *Homo*—that is, the novel emergence of the core components and their relationships, and thus of a new biologically significant whole. The usual contrast class is other primates: on this view, childhood is not found in other primate taxa and it would be a mistake to regard childhood as a mere modification in the timing of preexisting primate life-history stages. Rather, childhood would constitute a *de novo* trait (see Bogin 1997). The second claim concerns the derived character state of particular components, such as prolonged growth. But childhood in our species is multifaceted: we indeed spend a lot of time growing, and we do so in heavily structured systems of cooperative care, nourishing us not only physically but also informationally, supporting the learning of language, norms, and social and technical skills.

If it is not clear which traits are essential to individuate childhood and which are nonessential associated traits, we may incur two mistakes: (i) exaggerating the uniqueness of human childhood and (ii) undermining the possibility of justifiably recognizing childhood in other species. Indeed, a taxonomically biased characterization of childhood could hinder comparative analysis from the outset.

Several accounts of childhood place a premium on altriciality—that is, being born highly underdeveloped and completely dependent on parental care and the support of other members of the social group. Meerkats (*Suricata suricatta*) are an example of an altricial

social mammal. Pups are born with their eyes and ears closed and emit begging calls for up to three months to elicit food, modulating pitch to signal when they are ready for more complex prey. Meerkats also exhibit complex cooperative behavior in rearing their young. For instance, to teach pups how to handle prey while minimizing potential harm, adults typically disable or kill prey, such as by removing the sting from a scorpion, before offering them to the pups. The introduction of live prey occurs gradually and has been interpreted as a form of teaching (Hoppitt et al. 2012).

Thus, meerkats are altricial, undergo a developmental phase of dependency supported by the social group, and engage in social learning. Do meerkats, then, exhibit childhoods? Their dependent phase is not exceptionally prolonged (although not trivial for their lifespan), and the mechanisms of their learning and the use of information to solve a task differ from those of the human lineage. Whether these differences are sufficient grounds for denying the attribution of childhood beyond *Homo* hinges precisely on how we individuate the composite.

A second challenge is that, even if the core component traits of the composite are justifiably individuated in another taxon, there is still the question of the relationship among the components. Are the components working together in a way that justifies identifying the conjunction of underlying traits not just as a composite, but as the *same* composite? What requires comparison, then, is not merely the presence of the same components, but the patterned interactions that integrate them into a functionally and evolutionarily meaningful unit.

Components may co-occur across taxa yet stand in different developmental, functional, or selective relations to one another. If so, it becomes unclear whether we are dealing with the same composite trait or a superficially similar assemblage. The locus of disagreement would appear to be with the permissiveness of criteria. What kinds of relationships must components have in order for them to count as composites? And given a specific composite identified in one taxon, what criteria are sufficient for identifying it in other taxa?

Some influential studies on language in a comparative context represent a case in point. One example is a proposed (but not uncontroversial) distinction between a faculty of language “in a broad sense” (with traits pertaining to a sensory-motor system that supports speech, and others to the conceptual-perceptual system, sustaining semantics) and a faculty of language “in a narrow sense” (traits pertaining to the computational system, such as recursion and other syntactical features, see Hauser, Chomsky and Fitch 2002). Even when explicit proposals of how to delineate core components from additional ones have been put forth, making it an open empirical question which species have language (and when language first appeared in our lineage) faces the additional challenge of articulating the relationships among components, so that the comparable biological significance of the composite can be assessed.

6. Conclusions

Composite traits are common in biology and take a variety of forms, from morphological structures to behaviors to life-history stages. Composites are important to recognize as such,

since they carry biological significance transcending the significance of their constituent traits.

And with this higher-level significance come novel ecological and evolutionary roles.

At the same time, composite traits raise distinctive problems. Tracking a composite trait over time, across taxa, or between populations requires more than establishing the sameness of its core components. It requires justifying the components as essential to the composite, and it demands identifying the relationship the components must bear to one another in order for the composite to manifest. This is because the identity of a composite is not secured solely by the presence of components, but by the structured integration that renders the whole biologically significant.

In this paper, we have argued that identifying traits as composites is crucial for adequately characterizing, measuring and explaining variation in biology and anthropology. Our argument is a first step toward the development of a framework for composite traits individuation, one that takes seriously their explanatory indispensability and biological complexity.

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